

Enhancement of scale factor control accuracy of modulation algorithm for a closed-loop fiber-optic gyroscope.

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May 19, 2026

Abstract

TODO abstract

1 Motivation

iFOG principle of operation is based around the Sagnac effect. Its design is known for many years already, but as technology evolves the limits holding development are pushed forward, enabling for better precision, miniaturized size or lower power consumption. At the same time, the ever progressing industry finds new and more strict applications for the gyroscopes, setting the requirements bar even higher. Such context makes the already exercised designs being tested and optimized in every possible aspect. This applies equally to all the optical and electro-optic components, the electronics and the driving software. Crucial part of the control system of such gyroscope is the modulator. In a closed-loop design it is a vital element of the whole feedback loop, which directly affects the overall performance. It drives the optical circuit into a predefined state, in which the circuit response can be demodulated and error signals be determined. Modulator accuracy has critical impact on the resulting performance. This paper takes a closer look at the all-digital implementation of the modulator subsystem and particularly on the algorithm controlling it. Mathematical analysis is conducted to allow for pure software enhancements in the algorithm accuracy. Then, by placing the algorithm in the context of various hardware configurations, its accuracy is also determined for each.

2 Principle of operation - open-loop configuration

A fibre-optic gyroscope utilizes Sagnac effect. It generates a light beam, which is ideally split in 50/50 ratio. Each half then enters a loop of relatively long fiber through each of its ends. The fiber length L determines light transit time τ through it, in both directions (Equation 1, where n is the refraction coefficient, affecting the speed of light in the fiber medium).

$$\tau = \frac{L}{n \cdot c} \quad (1)$$

The halves go past each other and exit on the opposite ends. The optical circuit is arranged, so that the two exiting beams interfere with each other and the resulting light intensity is measured at a detector. When the circuit stays at rest, the beams interfere with zero phase shift ($\Delta\phi$), resulting in the maximum intensity P_0 measured at the detector. However, when the circuit rotates, the two halves travel paths of slightly different lengths, resulting in a non-zero phase shift. The intensity measured at the detector is phase-shift dependent:

$$\Delta\phi = \frac{L \cdot D \cdot \omega}{n \cdot c \cdot \lambda} \quad (2)$$

where D - fiber loop diameter, ω - angular velocity, λ - wavelength. This effect has been depicted in Figure 1.

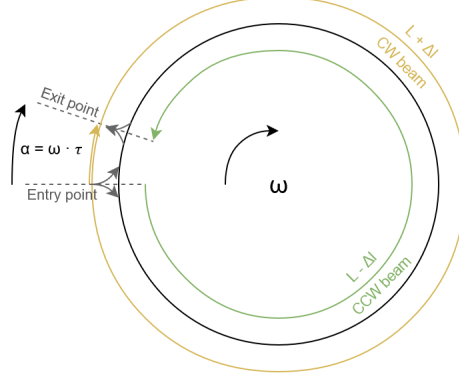


Figure 1: Two counterpropagating light beams enter the fiber loop at the *Entry point*. Their phase is initially in sync. After the transit time τ they exit and interfere. However, during that time, the *Entry/Exit point* has moved according to rotation rate ω (by the angle $\alpha = \omega \cdot \tau$) and the CW and CCW beams covered non-equal path lengths ($L \pm \Delta l$). In result they differ in phase.

The formula in Equation 3 is the optical circuit response. It describes how the measured light intensity depends on the phase shift, while phase shift itself results from Equation 2.

$$P(\Delta\phi) = \frac{P_0}{2} \cdot (1 + \cos(\Delta\phi)) \quad (3)$$

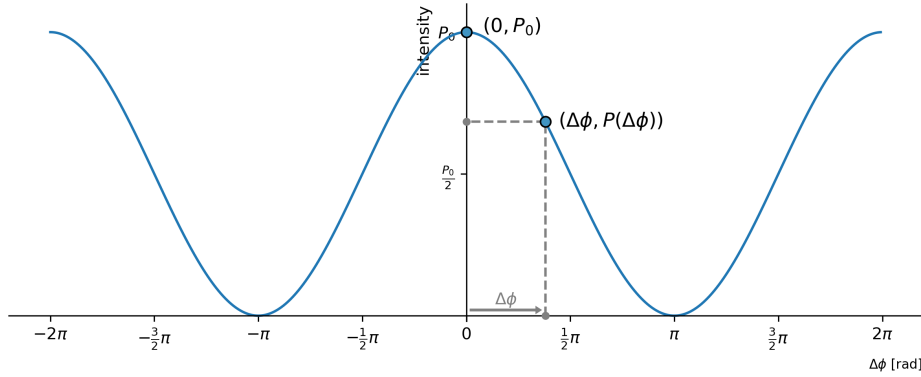


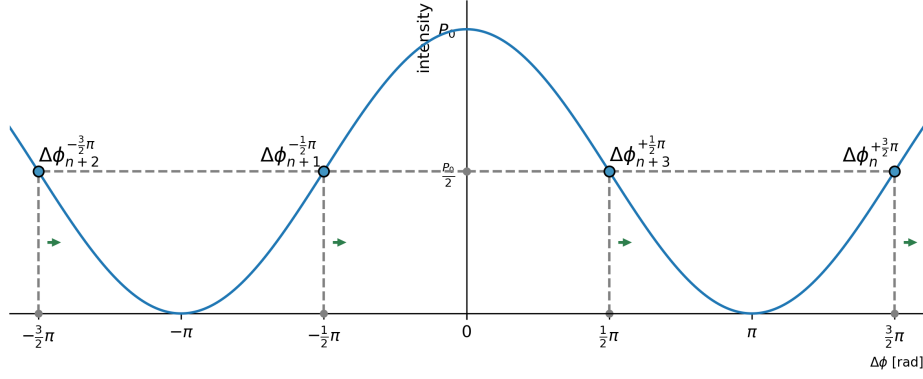
Figure 2: Optical circuit response function is a shifted cosine. The two exemplary operating points refer to the circuit being at rest - in $(0, P_0)$ - and subjected to rotation, which results in non-zero phase shift - in $(\Delta\phi, P(\Delta\phi))$.

When the circuit is at rest ($\omega = 0 \rightarrow \Delta\phi = 0$), the operating point resides at peak intensity P_0 . A non-zero rotation results in a phase shift which moves the operating point according to the rate and direction, most likely decreasing the measured intensity. Two problems are linked with such open-loop operation. Firstly, the intensity drop does not allow to tell the direction of rotation. Secondly, in the resting region, the slope is flat, thus the sensitivity is zero. With greater rotation rate, the slope becomes steep, reaching maximum sensitivity in $\frac{\pi}{2}$, but then decreases again. For these reasons a closed-loop configuration will be superior in performance.

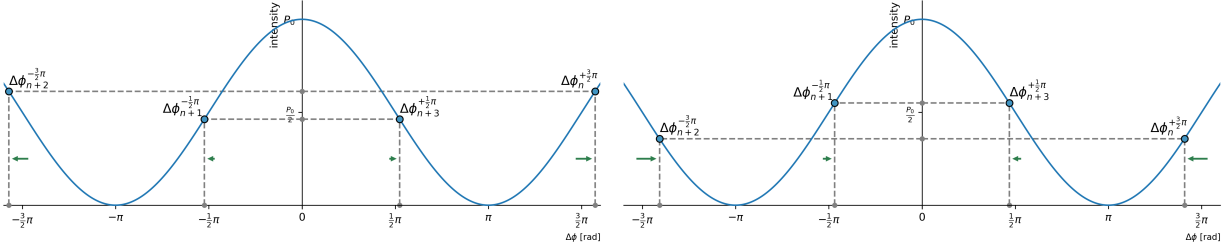
3 Modulation scheme - closed-loop configuration

In closed-loop configuration of an iFOG, the control system biases optical circuit sequentially in four points - i.e. $\Delta\phi = +\frac{3}{2}\pi, -\frac{1}{2}\pi, -\frac{3}{2}\pi, +\frac{1}{2}\pi$ - utilizing the *four-step modulation*. This allows for linearizing the response,

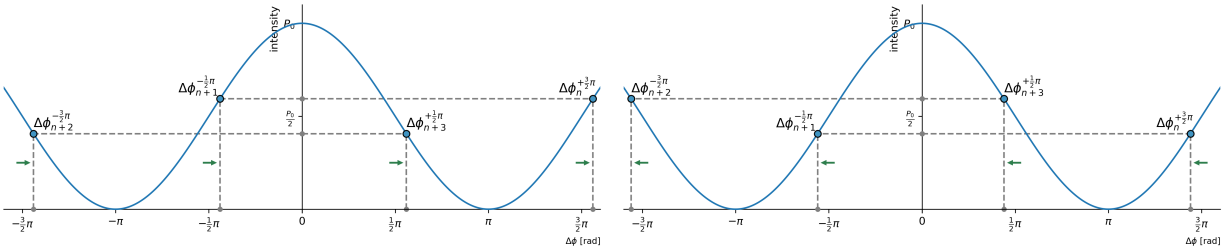
keeping it at maximum sensitivity and detecting the rotation direction. Additionally, compared to *two-step modulation* this scheme is supreme, as it allows for controlling the scale factor. This is crucial due to several reasons. The applied phase shift is controlled by an electronic circuit. It performs the calculations digitally and applies a corresponding voltage using digital-to-analog converter and an electronic front-end. The voltage then gets translated to a phase shift value in an electro-optic modulator device. All of these are subjects temperature drift and aging, which affect the combined gain. For this reason constant calibration is desired for maintaining measurement accuracy. Diagrams in Figure 3 depict how the biased circuit responds to various conditions - perfectly biased, scale factor error and rotation present.



(a) Ideal case where the scale factor is perfectly adjusted and rotation induced phase shift is either zero or actively counteracted by the device.



(b) Scale factor too high - bias points are shifted away from zero, resulting in intensity imbalance. (c) Scale factor too low - bias points shifted towards zero, resulting in opposite imbalance.



(d) Phase shift induced by positive rotation - bias points shifted right. Odd bias points correspond to lower intensity than even bias points. (e) Phase shift induced by negative rotation - bias points shifted left. Odd bias points correspond to higher intensity than even bias points.

Figure 3: Four-step modulation allows for detection of scale factor error and Sagnac-induced phase shift.

Control system of the device biases the circuit continuously in the following step order $\{\Delta\phi_n^{+\frac{3}{2}\pi}, \Delta\phi_{n+1}^{-\frac{1}{2}\pi}, \Delta\phi_{n+2}^{-\frac{3}{2}\pi}, \Delta\phi_{n+3}^{+\frac{1}{2}\pi}\}$. The modulation frequency corresponds to twice the loop transit time. In every cycle, the software monitors the scale factor error and adjusts its value accordingly. Secondly, it monitors the rotation induced phase shift, for which it counteracts, by applying additional phase shift of the same amplitude and opposite sign. This way it nulls the effective shift. by knowing the amount of nullled shift, it can compute the observed rotation rate.

4 Hardware architecture

4.1 Classic architecture - Dedicated gain stage

4.2 Thin architecture - All-digital approach

5 Algorithm requirements and design

6 Scale factor error

TODO

Scale factor error has two sources. Primarily the algorithm needs to adjust the numerical value precisely to control the scale factor. Error in calculating this value will directly impact the accuracy. However, even with a perfect algorithm the quantization noise resulting from the usage of a DAC leaves some error preset at all times. While the first factor is a subject for algorithmic optimization, the second one results from the physical circuit implementation and depends on several following parameters.

Firstly, the relative scale factor error can be denoted as in Equation 4.

$$\delta(n) = \frac{1}{2} \cdot \frac{\Delta v}{V_{\pi}(n)} \quad (4)$$

Voltage resolution of the used DAC results directly from the number of bits of the converter (Equation 5).

$$\Delta v = V_{ref} \cdot 2^{-N} \quad (5)$$

Minimal DAC reference voltage shall to be adjusted to cover all the required operating span (Equation 6).

$$V_{ref}^{min} = V_{\pi}^{max} \cdot R \cdot M \quad (6)$$

where range $R = 3.5$ results directly from the algorithm implementation (it shall be able to drive the values equivalent to $\pm 1\frac{3}{4}\pi$, yielding the span of $R = 3.5$) and margin $M \in [1.0; 1.1]$. When determining the final reference voltage it is advantageous to align it with a voltage multiple of one of typically available chips. It makes implementing the electronic circuit more straightforward, thus eliminating possible noise sources. The resulting reference voltage will in most cases be slightly higher than the minimal value (Equation 7). In practice, the best compromise is to utilize this excess for the margin M , so that after adding it, the $V_{ref} \approx V_{ref}^{min}$.

$$V_{ref} = \lceil V_{ref}^{min} \rceil \quad (7)$$

Substituting TODO yields the formula for relative scale factor error in a worst case scenario, when the real scale factor remains exactly in between the two controllable, discrete phase shift values, but with the assumption, that the software has perfectly calculated the scale (no error introduced by the algorithm).

$$\delta(n) = \frac{1}{2} \cdot \frac{\lceil V_{\pi}^{max} \cdot R \cdot M \rceil \cdot 2^{-N}}{V_{\pi}(n)} \quad (8)$$

An exemplary calculation assuming some real-world values:

$$\begin{aligned} \delta_{ex} &= \frac{1}{2} \cdot \frac{\lceil 4.5V \cdot 3.5 \cdot 1.04 \rceil \cdot 2^{-16}}{3.5V} = \frac{\lceil 16.38V \rceil \cdot 2^{-16}}{7V} \\ &= \frac{\overbrace{16.384V}^{4.4,096V} \cdot 2^{-16}}{7V} = 2^{-16} \cdot 2.34 = 2^{-16} \cdot 2^{1.23} \end{aligned} \quad (9)$$

$$\begin{aligned}
\phi_{n-2}^{-\frac{3}{2}\pi} &= -\frac{3}{4}\pi \cdot (1 + \delta_{n-2}) \\
\phi_{n-1}^{+\frac{1}{2}\pi} &= +\frac{1}{4}\pi \cdot (1 + \delta_{n-1}) \\
\phi_{n+0}^{+\frac{3}{2}\pi} &= +\frac{3}{4}\pi \cdot (1 + \delta_{n+0}) \\
\phi_{n+1}^{-\frac{1}{2}\pi} &= -\frac{1}{4}\pi \cdot (1 + \delta_{n+1}) \\
\phi_{n+2}^{-\frac{3}{2}\pi} &= -\frac{3}{4}\pi \cdot (1 + \delta_{n+2}) \\
\phi_{n+3}^{+\frac{1}{2}\pi} &= +\frac{1}{4}\pi \cdot (1 + \delta_{n+3})
\end{aligned} \tag{10}$$

6.1 Scale factor error feedback

$$\Delta\phi_n = (\phi_n - \phi_{n-2}) + \Delta\phi_n^{sagnac} \tag{11}$$

Assuming no Sagnac effect:

$$\begin{aligned}
P_n^{+\frac{3}{2}\pi} &= \frac{P_0}{2} \cdot (1 + \cos(\Delta\phi_n)) = \frac{P_0}{2} \cdot (1 + \cos(\phi_n - \phi_{n-2})) = \\
&= \frac{P_0}{2} \cdot (1 + \cos(+\frac{3}{2}\pi \cdot (1 + \frac{\delta_n + \delta_{n-2}}{2}))) \\
&= \frac{P_0}{2} \cdot (1 + \frac{\delta_n + \delta_{n-2}}{2}) \\
P_{n+1}^{-\frac{1}{2}\pi} &= \frac{P_0}{2} \cdot (1 + \frac{\delta_{n+1} + \delta_{n-1}}{2}) \\
P_{n+2}^{-\frac{3}{2}\pi} &= \frac{P_0}{2} \cdot (1 + \frac{\delta_{n+2} + \delta_n}{2}) \\
P_{n+3}^{+\frac{1}{2}\pi} &= \frac{P_0}{2} \cdot (1 + \frac{\delta_{n+3} + \delta_{n+1}}{2})
\end{aligned} \tag{12}$$

$$\begin{aligned}
P(\phi_B + \delta) &= \frac{P_0}{2} \cdot (1 + \cos(\phi_B + \delta)) = \\
&= \frac{P_0}{2} \cdot (1 + \underbrace{\cos(\phi_B)}_0 + \cos'(\phi_B) \cdot \delta) = \\
&= \frac{P_0}{2} \cdot (1 - \sin(\phi_B) \cdot \delta)
\end{aligned} \tag{13}$$

where $\phi_B \in \{+\frac{3}{2}\pi, +\frac{1}{2}\pi, -\frac{1}{2}\pi, -\frac{3}{2}\pi\}$.

$$\begin{aligned}
e_{V_\pi}^{-\frac{1}{2}\pi}(n+1) &= P(n) - P(n+1) = \\
&= \frac{P_0}{2} \cdot (\frac{\delta_n + \delta_{n-2}}{2} - \frac{\delta_{n+1} + \delta_{n-1}}{2}) = \\
&= \frac{P_0}{4} \cdot (\underbrace{\delta_{n-2} - \delta_{n-1}}_{\text{last two inputs}} + \underbrace{\delta_n - \delta_{n+1}}_{\text{current two inputs}}) \\
e_{V_\pi}^{+\frac{1}{2}\pi}(n+3) &= P(n+2) - P(n+3) = \\
&= \frac{P_0}{2} \cdot (\frac{\delta_{n+2} + \delta_n}{2} - \frac{\delta_{n+3} + \delta_{n+1}}{2}) \\
&= \frac{P_0}{4} \cdot (\underbrace{\delta_n - \delta_{n+1}}_{\text{last two inputs}} + \underbrace{\delta_{n+2} - \delta_{n+3}}_{\text{current two inputs}})
\end{aligned} \tag{14}$$

References

[Doe] *First and last L^AT_EX example.*, John Doe 50 B.C.